

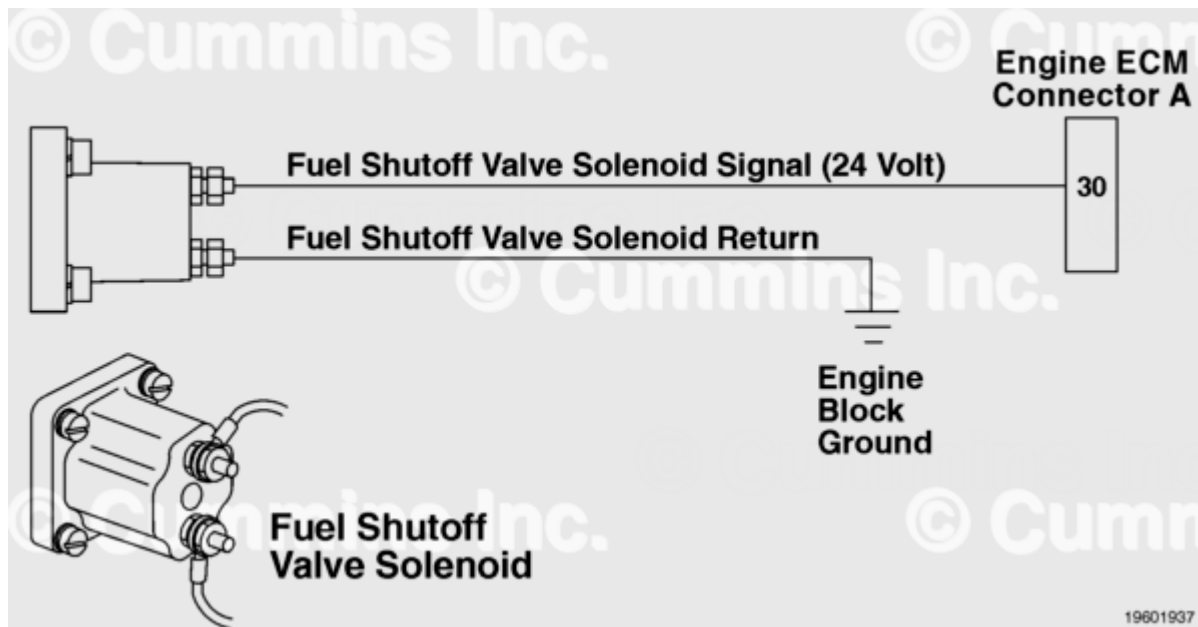
Fault Code: 254

Fuel Shutoff Solenoid Supply Circuit

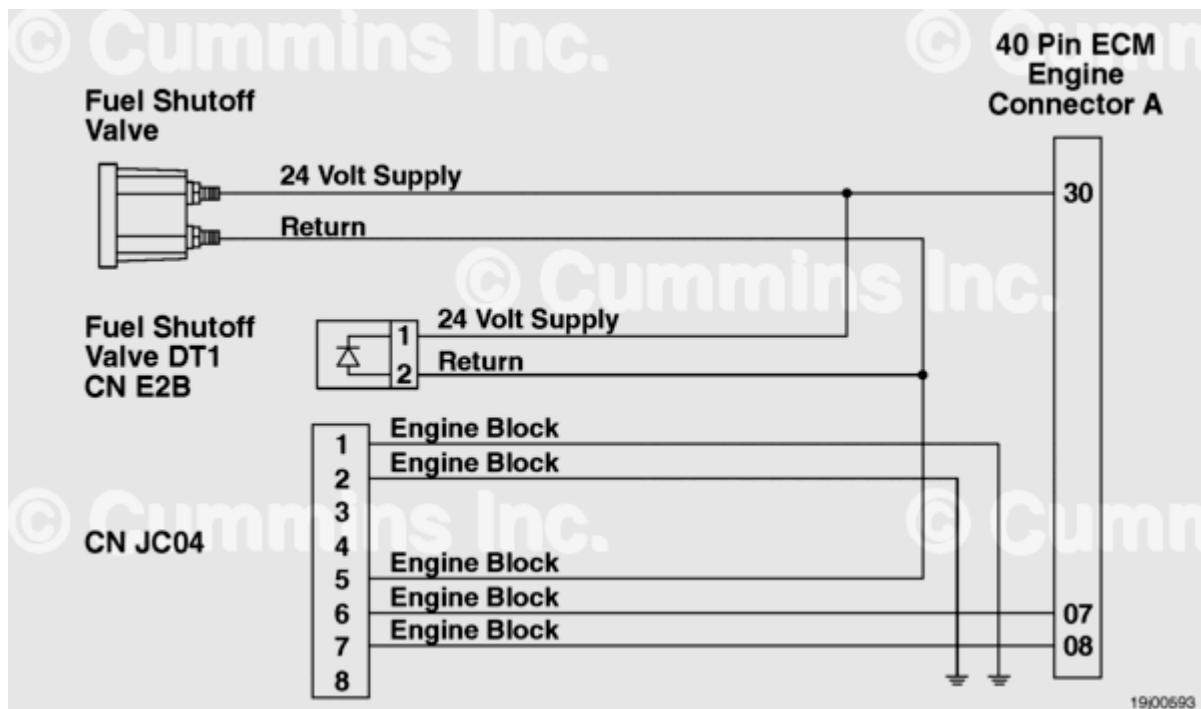
 Printable Version

Overview

Codes	Reason	Effect
Fault Code: 254 PID(P): S17 SPN: 632 FMI: 4 Lamp: None SRT: 00-365	The fuel shutoff valve solenoid circuit is open or shorted. Less than 6.0 VDC detected at fuel shutoff valve solenoid supply pin 30 of the engine harness or resistance of the solenoid has dropped below 20 ohms.	Engine control module (ECM) turns off fuel shutoff valve supply voltages. The engine dies.



QSK19, QSK45, and QSK78 - Fuel Shutoff Solenoid Circuit



QSK23 - Fuel Shutoff Solenoid Circuit

Circuit Description

The fuel shutoff solenoid is a device used by the ECM to control the fuel supply to the rail fueling circuit. The ECM can shut down the engine by cutting off the power to the fuel shutoff solenoid.

Component Location

The fuel shutoff solenoid is located between the rail and timing actuators on the control valve body behind the ECM.

Shoptalk

- The fuel shutoff solenoid **only** stops fuel to the rail circuit; the timing circuit bypasses the fuel shutoff solenoid.
- Inspect the fuel shutoff supply circuit for external wires that can be spliced in to power another device. Remove any extra wires that are found in the circuit.
- If there is an external shutdown system on the vehicle that uses the fuel shutoff valve for engine shutdown, make sure it has **not** malfunctioned and lowered the voltage on the fuel shutoff circuit.
- Inspect the engine block-to-chassis ground wire to make sure it is securely fastened to a clean, dry surface.
- Check the starter solenoid positive (+) terminal for a loose connector and/or accessory wiring with damaged insulation.
- The engine will have to be cranked before any voltage is present at the fuel shutoff valve terminals. Once the ECM has received the 50-rpm signal, the voltage will remain supplied to

the fuel shutoff valve until the keyswitch is cycled to the OFF position.

Note: This calibration feature is in effect for **only** QSK19 engines manufactured after September 1999, but is included for all QSK23, QSK45, QSK60, and QSK78 engines.

Refer to Troubleshooting Fault Code t05-254 (/qs3/pubsys2/xml/en/procedures/19/19-t05-254.html)

Last Modified: 02-Jan-2017
